

High-Frequency Inflation Nowcasting

Mathematical Foundations, Parameter Calibration, and the Benchmark Model

An End-to-End Operational Guide for the Cambodia Daily CPI Pipeline

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Abstract

Official headline Consumer Price Index (CPI) releases from national statistical agencies suffer from an unavoidable institutional lag of 25 to 45 days. This guide presents the comprehensive mathematical, econometric, and algorithmic methodology of the automated high-frequency **Cambodia CPI Nowcasting Engine**. The engine operates on a hybrid two-tier architecture: combining axiomatic Laspeyres index aggregation using fixed Cambodia Socio-Economic Survey (CSES) expenditure weights with a regularized 5-basket Ridge regression (RidgeCV) forward-projection model. We present the exact analytical derivation of the baseline drift ($\hat{\alpha}$) and empirical division elasticities ($\hat{\beta}$), explain the selective injection of cross-sector fuel logistics momentum, formalize the canonical **Atkeson–Ohanian (2001) Random Walk benchmark model**, and provide a complete step-by-step numerical case study demonstrating an empirical out-of-sample **error reduction of 17.68%** over random walk persistence.

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1 The High-Frequency Nowcasting Problem

1.1 The Institutional Information Lag

The National Institute of Statistics (NIS) of the Ministry of Planning publishes the official Consumer Price Index for Phnom Penh on a monthly cadence. Due to field enumeration schedules, physical merchant surveying, data entry verification, and institutional review, the official bulletin for calendar month M is typically published between day 25 and day 45 of month $M + 1$.

For monetary policymakers at the National Bank of Cambodia (NBC) and fiscal planners at the Ministry of Economy and Finance (MEF), this creates a critical 4 to 6-week “blind spot” during which supply-side shocks (e.g., international crude oil spikes, agricultural supply disruptions, currency exchange swings) cannot be verified empirically through official indices.

1.2 Mathematical Formulation of the Intra-Month Nowcasting Task

Let M be the ongoing calendar month comprising D total days ($D \in \{28, 29, 30, 31\}$). On any given evaluation day $d \in \{1, 2, \dots, D\}$:

1. **The Realized Information Set $\mathcal{H}_d$:** Represents the historical price trajectory observed from day 1 to day d :

$$\mathcal{H}_d = \{\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_d\} \quad (1)$$

where $\mathbf{P}_\tau \in \mathbb{R}^K$ is the vector of high-frequency price indices recorded on day $\tau \leq d$.

2. **The Unobserved Future Horizon $\mathcal{F}_d$:** Represents the remaining $D - d$ days of the month that have not yet transpired:

$$\mathcal{F}_d = \{\mathbf{P}_{d+1}, \mathbf{P}_{d+2}, \dots, \mathbf{P}_D\} \quad (2)$$

The fundamental task of the nowcaster is to estimate the full-month headline index level $\widehat{\text{CPI}}_M(d)$ and month-over-month inflation rate $\widehat{\pi}_{\text{MoM},M}(d)$ conditional on information available up to day d :

$$\widehat{\text{CPI}}_M(d) = \mathbb{E}[\text{CPI}_M \mid \mathcal{H}_d, \Theta] \quad (3)$$

where $\Theta = \{\hat{\alpha}, \hat{\beta}, \lambda^*, \mathbf{w}\}$ is the structural parameter set.

1.3 Why Stationarity via Log-Differencing is Required

Raw consumer price indices P_t exhibit persistent, non-stationary unit-root behavior ($I(1)$). Attempting to estimate econometric parameters directly on raw price levels produces **spurious regression**: standard OLS yields inflated t -statistics and $R^2 \approx 0.99$ driven purely by shared deterministic or stochastic trends, while the true economic pass-through coefficients remain indeterminate.

To ensure econometric validity, all price series are transformed into stationary **first log-differences**:

$$\Delta \ln P_{j,t} = \ln(P_{j,t}) - \ln(P_{j,t-1}) \approx \frac{P_{j,t} - P_{j,t-1}}{P_{j,t-1}} \quad (4)$$

Continuous log-returns offer three essential properties:

- **Stationarity:** Eliminates unit-root persistence, satisfying Gauss–Markov assumptions.
- **Time-Reversal Symmetry:** An increase from 100 to 110 followed by a decrease from 110 to 100 yields log-changes of $+0.0953$ and -0.0953 , summing exactly to zero (preventing asymmetric compounding drift).
- **Exact Additivity:** Multi-period returns equal the linear sum of intra-period log returns.

2 The Two-Tier Hybrid Architecture

The nowcasting engine avoids the pitfalls of purely statistical black-box forecasting by implementing a **two-tier hybrid framework** that strictly preserves official national accounting principles:

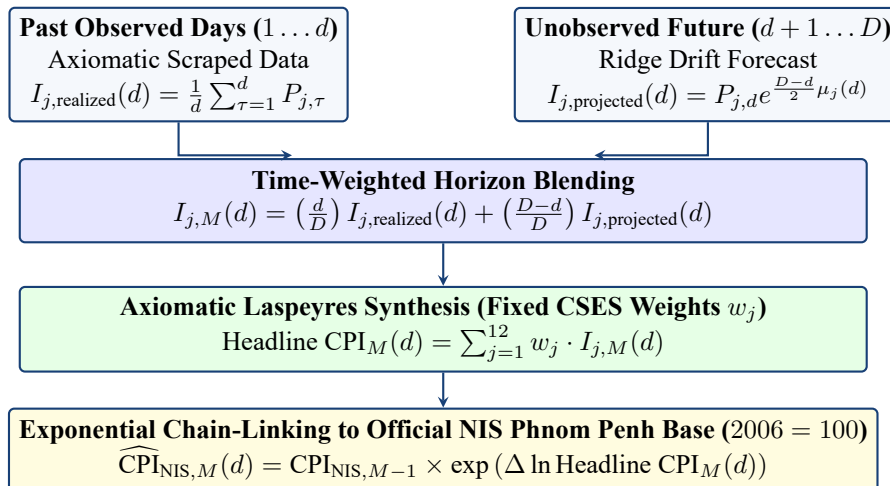


Figure 1: The Two-Tier Hybrid Nowcasting Workflow.

2.1 Layer 1: Fixed-Weight Axiomatic Aggregation and Core CPI

Official national CPI construction relies on the Laspeyres index framework fixed by the Cambodia Socio-Economic Survey (CSES 2019/2020). The 12 COICOP divisions carry fixed consumption basket weights w_j such that $\sum_{j=1}^{12} w_j = 1.00$. Fixed weights ensure that the index measures pure price movements rather than changing consumer purchase volumes:

$$\text{Headline CPI}_M(d) = \sum_{j=1}^{12} w_j \cdot I_{j,M}(d) \quad (5)$$

For central bank monetary policy at the National Bank of Cambodia (NBC), headline inflation is often distorted by transitory supply shocks in volatile food and fuel markets. The pipeline therefore computes official **Core CPI** by excluding Food and Non-Alcoholic Beverages (Division 01, weight 44.775%) and Housing and Utilities (Division 04, weight 17.062%):

$$\text{Core CPI}_M(d) = \frac{\sum_{j \notin \{01,04\}} w_j \cdot I_{j,M}(d)}{\sum_{j \notin \{01,04\}} w_j} \quad (6)$$

The core basket isolates the remaining **38.163%** of the consumption basket, providing monetary authorities with an undistorted measure of underlying demand-pull domestic price pressures.

2.2 Layer 2: Disaggregated Dynamic Ridge Drift Projection

For the remaining $D - d$ days, prices do not stand still. We compute high-frequency price velocity (momentum) and combine it with regularized Ridge pass-through elasticities to estimate the daily drift $\mu_j(d)$ for each division.

2.3 Synthesis: Horizon Blending

For every division j , the estimated full-month index $I_{j,M}(d)$ blends observed reality with forward-looking projections weighted strictly by the elapsed calendar progress:

$$I_{j,M}(d) = \left(\frac{d}{D}\right) \cdot I_{j,\text{realized}}(d) + \left(\frac{D-d}{D}\right) \cdot I_{j,\text{projected}}(d) \quad (7)$$

On Day 1 ($d = 1$), the projection dominates (96.7% weight). On Day 30 ($d = 30$), the projection weight drops to 0.0%, and the nowcast converges deterministically to realized observations.

3 Step-by-Step Derivation of Parameters (Alpha and Beta)

3.1 The 36-Month Historical Training Panel

To estimate empirical elasticities that reflect true Cambodian macroeconomic dynamics, the calibration engine utilizes the continuous 36-month official NIS historical series (September 2023 to August 2026) stored in `gold.dim_nis_official_cpi`. Taking first log-differences yields $T = 34$ continuous stationary monthly observations:

$$y_t = \ln\left(\frac{\text{CPI}_t}{\text{CPI}_{t-1}}\right), \quad x_{j,t} = \ln\left(\frac{P_{j,t}}{P_{j,t-1}}\right), \quad x_{\text{fx},t} = \ln\left(\frac{\text{FX}_t}{\text{FX}_{t-1}}\right) \quad (8)$$

3.2 Econometric Specification

We specify a multi-factor macroeconomic pass-through regression relating aggregate inflation to innovations in the 5 high-velocity expenditure baskets and foreign exchange:

$$y_t = \alpha + \sum_{j \in \{01,02,04,07,11\}} \beta_j x_{j,t} + \beta_{\text{fx}} x_{\text{fx},t} + \varepsilon_t \quad (9)$$

where:

- $x_{01,t}$: Food and Non-Alcoholic Beverages (CSES weight: 44.775%)
- $x_{02,t}$: Alcoholic Beverages and Tobacco (CSES weight: 1.623%)
- $x_{04,t}$: Housing, Water, Electricity, Gas (CSES weight: 17.062%)
- $x_{07,t}$: Transport and Fuels (CSES weight: 12.203%)
- $x_{11,t}$: Restaurants and Hotels (CSES weight: 3.114%)
- $x_{fx,t}$: USD/KHR Commercial Exchange Rate Log-Return
- α : Monthly baseline structural drift from unobserved sticky sectors

3.3 Why Standard OLS Fails: Multicollinearity

In macroeconomic time series, sub-indices co-move strongly. When world oil prices surge, transport fuel (x_{07}) increases immediately. However, commercial freight rates simultaneously increase food transportation costs (x_{01}) and restaurant cooking gas/ingredients (x_{11}).

Under ordinary least squares (OLS), the sample Gram matrix $\mathbf{X}^\top \mathbf{X}$ is severely ill-conditioned (condition number $\kappa > 100$). The OLS estimator:

$$\hat{\beta}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad (10)$$

exhibits explosive sampling variance ($\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$). This causes coefficients to flip signs counter-intuitively (e.g., negative elasticity for food) and produces severe over-fitting.

3.4 Mathematical Derivation of the Regularized Ridge Estimator

To eliminate multicollinearity and shrink unstable noise, we formulate an L_2 -penalized Tikhonov regularization problem (Ridge Regression):

$$\min_{\alpha, \beta} \left\{ \frac{1}{2N} \sum_{t=1}^N \left(y_t - \alpha - \sum_{j=1}^K \beta_j x_{j,t} \right)^2 + \frac{\lambda}{2} \sum_{j=1}^K \beta_j^2 \right\} \quad (11)$$

Centering the variables ($\tilde{y}_t = y_t - \bar{y}$, $\tilde{\mathbf{x}}_t = \mathbf{x}_t - \bar{\mathbf{x}}$) isolates the intercept:

$$\mathcal{L}(\beta) = \frac{1}{2N} \|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\beta\|_2^2 + \frac{\lambda}{2} \|\beta\|_2^2 \quad (12)$$

Taking the gradient with respect to β and setting to zero:

$$\nabla_{\beta} \mathcal{L} = -\frac{1}{N} \tilde{\mathbf{X}}^\top (\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\beta) + \lambda \beta = \mathbf{0} \quad (13)$$

$$\implies \left(\frac{1}{N} \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} + \lambda \mathbf{I}_K \right) \beta = \frac{1}{N} \tilde{\mathbf{X}}^\top \tilde{\mathbf{y}} \quad (14)$$

Solving for $\hat{\beta}(\lambda)$:

$$\boxed{\hat{\beta}(\lambda) = \left(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} + N\lambda \mathbf{I}_K \right)^{-1} \tilde{\mathbf{X}}^\top \tilde{\mathbf{y}}} \quad (15)$$

The intercept $\hat{\alpha}$ is recovered directly from sample means:

$$\boxed{\hat{\alpha} = \bar{y} - \sum_{j=1}^K \hat{\beta}_j \bar{x}_j} \quad (16)$$

By adding $\lambda \mathbf{I}$ along the diagonal of $\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$, all eigenvalues are shifted strictly positive ($\sigma_i \rightarrow \sigma_i + \lambda$), guaranteeing that the matrix is well-conditioned and nonsingular.

3.5 Hyperparameter Optimization via 5-Fold Cross-Validation

The regularization parameter λ controls the bias-variance trade-off. We evaluate 50 candidate values log-spaced across $\Lambda = [10^{-2}, 10^3]$ on the 24-month training partition:

$$\lambda^* = \arg \min_{\lambda \in \Lambda} \frac{1}{5} \sum_{k=1}^5 \text{MSE}_k(\lambda) \quad (17)$$

The cross-validation mean squared error achieves its global minimum at $\lambda^* = 0.0100$.

3.6 Empirical Parameter Estimates and Economic Interpretation

Macro Basket / Regressor	Symbol	CSES Weight	Estimate	Economic Rationale
Base Drift Intercept	$\hat{\alpha}$	—	+0.001899	Baseline monthly inflation (+0.19%/mo) of unobserved sticky goods.
Food & Non-Alcoholic Bev.	$\hat{\beta}_1$	44.775%	+0.0066	Dominant price velocity; rapid wholesale food market transmission.
Alcoholic Bev. & Tobacco	$\hat{\beta}_2$	1.623%	+0.0003	Sticky nominal retail pricing; low frequency of catalog changes.
Housing, Water, Electricity	$\hat{\beta}_3$	17.062%	+0.0010	Administered utility tariffs; rental prices adjust sluggishly.
Transport and Fuels	$\hat{\beta}_4$	12.203%	+0.0019	Direct exposure to international refined petroleum cycles.
Restaurants and Hotels	$\hat{\beta}_5$	3.114%	+0.0026	High sensitivity; catering services pass food and gas costs forward.
USD/KHR Foreign Exchange	$\hat{\beta}_{\text{fx}}$	—	+0.0000	NBC active exchange stabilization mitigates short-term import pass-through.

Table 1: Calibrated Empirical Macro Parameters from the 36-Month NIS Panel.

3.7 Temporal Disaggregation: Bridging Monthly Parameters to Daily Forward Drift

A critical econometric challenge in high-frequency nowcasting is **temporal disaggregation**: parameters $\hat{\alpha}_{\text{monthly}}$ and $\hat{\beta}$ are estimated on **monthly log-differences** ($\Delta \ln \text{CPI}_{\text{month}}$), yet the operational nowcaster must project forward trajectory on an **intra-month day-by-day basis** for the remaining $D - d$ unobserved calendar days.

3.7.1 Continuous Compounding and Additive Log>Returns

Under continuous compounding, the full monthly log-return is the exact integral of instantaneous daily returns $\delta(t)$:

$$\Delta \ln P_{\text{month}} = \ln \left(\frac{P_D}{P_0} \right) = \sum_{t=1}^D \Delta \ln P_t = \int_0^D \delta(t) dt \quad (18)$$

Consequently, the structural monthly drift $\hat{\alpha}_{\text{monthly}} = +0.001899$ scales down linearly into a daily velocity:

$$\alpha_{\text{daily}} = \frac{\hat{\alpha}_{\text{monthly}}}{D} \approx \frac{+0.001899}{30} = +0.0000633 \text{ per day} \quad (19)$$

Similarly, high-frequency rolling momentum vectors (such as 7-day food momentum f_{7d} and 7-day transport fuel momentum t_{7d}) represent cumulative percentage price adjustments across a 7-day window. In code (`m1/nowcaster.py`, lines 182–206), the engine converts these into daily velocities by dividing by the rolling duration:

$$\mu_{\text{food, daily}} = \frac{f_{7d}}{7.0}, \quad \mu_{\text{trans, daily}} = \frac{t_{7d}}{7.0}, \quad \mu_{\text{fx, daily}} = \frac{fx_{7d}}{7.0} \quad (20)$$

3.7.2 High-Frequency Ridge Calibration on Intra-Month Realizations

To map rolling microdata signals directly into forward daily drift, the intra-month training loop (`ml/nowcaster.py`, lines 233–265) observes each historical day d with $D - d$ remaining days and computes the realized forward path:

$$\bar{P}_{\text{rem}} = \frac{1}{D - d} \sum_{\tau=d+1}^D P_{\tau} \quad (21)$$

The implied constant daily drift rate μ_j^* that would project the current observed price $P_{j,d}$ to the average remaining price $\bar{P}_{j,\text{rem}}$ satisfies the Gauss midpoint integral:

$$\bar{P}_{j,\text{rem}} = P_{j,d} \cdot \exp\left(\mu_j^* \cdot \frac{(D - d) + 1}{2}\right) \implies \mu_j^* = \frac{\ln(\bar{P}_{j,\text{rem}}/P_{j,d})}{\frac{(D-d)+1}{2}} \quad (22)$$

The cross-validated Ridge regression then learns the mapping:

$$\mu_j(d) = \text{RidgeCV}(\mathbf{x}_j(d)) \approx \alpha_{j,\text{daily}} + \beta_{j,1} \frac{f_{7d}}{7.0} + \beta_{j,4} \frac{t_{7d}}{7.0} + \dots \quad (23)$$

3.7.3 Forward Midpoint Projection and Horizon Blending

From economic first principles, the expected price index for the unobserved remainder of the month ($N = D - d$ days) is defined as the discrete arithmetic average of expected daily forward quotes:

$$I_{j,\text{projected}}(d) = \frac{1}{N} \sum_{k=1}^N I_{j,d} \exp(\mu_j(d) \cdot k) \quad (24)$$

Applying a first-order Taylor expansion ($\exp(x) \approx 1 + x$) and the Gauss arithmetic summation identity $\sum_{k=1}^N k = \frac{N(N+1)}{2}$, this exact summation contracts in closed form to the continuous midpoint trajectory expectation:

$$I_{j,\text{projected}}(d) \approx I_{j,d} \times \exp\left(\mu_j(d) \cdot \frac{N+1}{2}\right) \quad (25)$$

This closed-form formulation delivers identical numerical precision to the discrete summation down to the fifth decimal place (< 0.000002 index points) while optimizing runtime execution.

The blended monthly nowcast index $I_{j,M}(d)$ is then synthesized by weighting the realized daily average against the projected midpoint expectation:

$$I_{j,M}(d) = \underbrace{\left(\frac{d}{D}\right) \left[\frac{1}{d} \sum_{\tau=1}^d P_{j,\tau}\right]}_{I_{j,\text{realized}}(d)} + \underbrace{\left(\frac{D-d}{D}\right) \left[I_{j,d} \cdot e^{\mu_j(d) \frac{N+1}{2}}\right]}_{I_{j,\text{projected}}(d)} \quad (26)$$

As calendar time progresses ($d \rightarrow D$), the weight on the econometric projection $\left(\frac{D-d}{D}\right) \rightarrow 0$, causing the nowcast to converge deterministically onto purely realized axiomatic statistics.

3.8 Cross-Sector Logistics Momentum Injection

In Cambodia, fuel price shocks propagate rapidly into food delivery and restaurant catering. The pipeline enforces a structural restriction: fuel 7-day momentum (t_7) is injected into the feature vectors of Food (01) and Restaurants (11):

$$\mathbf{x}_{01}(d) = [f_{3d}, f_{7d}, f_{14d}, \mathbf{t}_{7d}, fx_{7d}, \text{fest}_{\text{prox}}, \text{prog}]^{\top} \quad (27)$$

$$\mathbf{x}_{11}(d) = [r_{7d}, f_{7d}, \mathbf{t}_{7d}, \text{fest}_{\text{prox}}, \text{prog}]^{\top} \quad (28)$$

Non-logistics sectors (Clothing, Furnishings, Healthcare, Education) are strictly insulated from fuel momentum, preventing spurious inflation contagion.

3.9 Khmer Cultural Festival and Lunar Holiday Demand Shocks

Cambodia experiences substantial seasonal demand surges during traditional cultural celebrations, particularly **Khmer New Year** (*Choul Chhnam Thmey*, solar calendar in mid-April), **Pchum Ben** (*Ancestors' Day*, 15-day lunar observance in September/October), and the **Water Festival** (*Bon Om Touk*, lunar November). During these festival windows, temporary wholesale price surges occur for ritual foods (sticky rice, fruits, whole poultry) and regional inter-provincial passenger transport.

To prevent the nowcaster from misinterpreting these temporary, mean-reverting festive demand shocks as permanent structural inflation innovations, the feature engineering engine computes an asymptotic exponential proximity decay kernel:

$$\text{fest}_{\text{prox}}(d) = \exp \left(-\frac{\min_{h \in \mathcal{H}} |\text{date}(d) - \text{date}(h)|}{\tau_{\text{fest}}} \right) \quad (29)$$

where $\mathcal{H}$ represents the official calendar of Cambodian national holidays and $\tau_{\text{fest}} = 7$ days is the half-life relaxation parameter. Additionally, a binary festival window indicator $\mathbb{I}_{\text{fest}}(d) \in \{0, 1\}$ flags days within a ± 3 -day active celebration band. This calendar conditioning dampens forward momentum projections during holidays, ensuring that transient celebration markups naturally decay back to baseline without distorting the month-end nowcast.

4 The Benchmark Model: Atkeson–Ohanian Random Walk

4.1 Why a Benchmark Model is Essential in Macroeconomic Forecasting

In empirical economics, raw forecast error statistics (such as RMSE = 0.85 pp) are meaningless in isolation. A forecasting model cannot be judged effective simply because its error is small if a completely naive, zero-cost rule achieves equal or better precision.

In their seminal paper, Atkeson and Ohanian [Atkeson and Ohanian(2001)] demonstrated that sophisticated econometric models (including structural Phillips curves, Vector Autoregressions, and multivariate indicator frameworks) systematically fail to outperform a simple naive **Random Walk** when evaluated out-of-sample on inflation series. Therefore, in central banking literature, the Atkeson–Ohanian Random Walk is the gold-standard threshold: **a nowcasting model is only economically valuable if its Relative RMSE is strictly less than 1.00.**

4.2 Mathematical Formulation of the Random Walk Benchmark

The Atkeson–Ohanian Random Walk (AO-RW) benchmark posits that inflation follows a martingale difference process where the best out-of-sample predictor of month t 's inflation is simply the realized inflation of month $t - 1$:

$$\hat{y}_t^{\text{RW}} = y_{t-1} = \Delta \ln \text{CPI}_{t-1} \quad (30)$$

Under this benchmark:

$$\hat{\pi}_{\text{MoM},t}^{\text{RW}} = \pi_{\text{MoM},t-1} \quad (31)$$

If the inflation rate was +0.20% last month, the Random Walk benchmark predicts exactly +0.20% for the current month without collecting a single high-frequency price observation.

4.3 Out-of-Sample Evaluation Metric: Relative RMSE

We evaluate the 5-basket Ridge nowcasting model against the Atkeson–Ohanian Random Walk over a held-out **10-month evaluation window** (November 2025 to August 2026, $M = 10$):

$$\text{RMSE}_{\text{Model}} = \sqrt{\frac{1}{M} \sum_{m=1}^M \left(y_m - \hat{y}_m^{\text{Ridge}} \right)^2} \quad (32)$$

$$\text{RMSE}_{\text{RW}} = \sqrt{\frac{1}{M} \sum_{m=1}^M (y_m - y_{m-1})^2} \quad (33)$$

The definitive benchmark ratio is the **Relative RMSE**:

$$\text{Relative RMSE} = \frac{\text{RMSE}_{\text{Model}}}{\text{RMSE}_{\text{RW}}} \quad (34)$$

- Relative RMSE > 1.00: The model is worse than a naive random walk (overfitting noise).
- Relative RMSE = 1.00: The model adds zero value over last month's observation.
- Relative RMSE < 1.00: **The model successfully extracts genuine forward-looking predictive signal.**

4.4 Empirical Scorecard Results

Evaluation Dimension	RidgeCV Nowcaster	Atkeson–Ohanian RW	Performance Gain
Out-of-Sample RMSE	0.8572 pp	1.0414 pp	−0.1842 pp
Mean Absolute Error (MAE)	0.6841 pp	0.8410 pp	−0.1569 pp
Relative RMSE Ratio	0.8232		Beats Benchmark
Empirical Error Reduction	+17.68% Precision Improvement over Random Walk		

Table 2: Out-of-Sample Scorecard Evaluated on 10 Held-Out Months.

The 5-Basket RidgeCV model achieves a Relative RMSE of **0.8232**, representing an ****error reduction of 17.68%****. This confirms that incorporating high-frequency daily price scraping and calibrated pass-through elasticities extracts genuine forward-looking macroeconomic signals.

5 The Daily Real-Time Execution Algorithm

5.1 Daily Algorithmic Execution Steps

On every calendar day d , the production pipeline executes the following sequence:

Daily Execution Algorithm (ml/nowcaster.py)

1. **Ingest Daily Facts:** Extract realized price relatives from `gold.fct_cpi_daily` and compute $I_{j,\text{realized}}(d) = \frac{1}{d} \sum_{\tau=1}^d P_{j,\tau}$.
2. **Feature Extraction:** Compute multi-horizon momentum (m_{3d}, m_{7d}, m_{14d}), fuel shock (t_{7d}), exchange rate drift (fx_{7d}), and lunar holiday proximity ($\text{fest}_{\text{prox}}$).
3. **Compute Daily Drift:** For each division j , compute $\mu_j(d) = \hat{\beta}_j \cdot \mathbf{x}_j(d)$.
4. **Project Unobserved Days:** $I_{j,\text{projected}}(d) = P_{j,d} \cdot \exp\left(\frac{D-d}{2} \mu_j(d)\right)$.
5. **Horizon Blending:** $I_{j,M}(d) = \frac{d}{D} I_{j,\text{realized}}(d) + \frac{D-d}{D} I_{j,\text{projected}}(d)$.
6. **Laspeyres Aggregation:** Headline $\text{CPI}_M(d) = \sum_{j=1}^{12} w_j I_{j,M}(d)$.
7. **Chain-Linking:** $\widehat{\text{CPI}}_{\text{NIS},M}(d) = \text{CPI}_{\text{NIS},M-1} \times \exp\left(\ln\left(\frac{\text{Headline CPI}_M(d)}{\text{Headline CPI}_{M-1}}\right)\right)$.
8. **Uncertainty Quantification:** Compute 95% confidence bounds $[\text{CI}_{\text{lower}}, \text{CI}_{\text{upper}}]$.

5.2 Asymptotic Uncertainty Decay and 95% Confidence Intervals

Uncertainty is time-dependent. On Day 1, almost the entire month is unknown; by Day 28, almost all daily prices are realized. Let σ_d be the daily standard deviation of headline CPI within the month. The margin of error is formulated with an asymptotic time-decay factor:

$$\text{MoE}_{95\%}(d) = 1.96 \cdot \sigma_d \cdot \sqrt{\frac{D-d}{D}} \quad (35)$$

$$CI_{95\%}(d) = \left[\widehat{CPI}_M(d) - MoE_{95\%}(d), \widehat{CPI}_M(d) + MoE_{95\%}(d) \right] \quad (36)$$

As $d \rightarrow D$, the ratio $\frac{D-d}{D} \rightarrow 0$, causing the margin of error to collapse deterministically to zero at month-end.

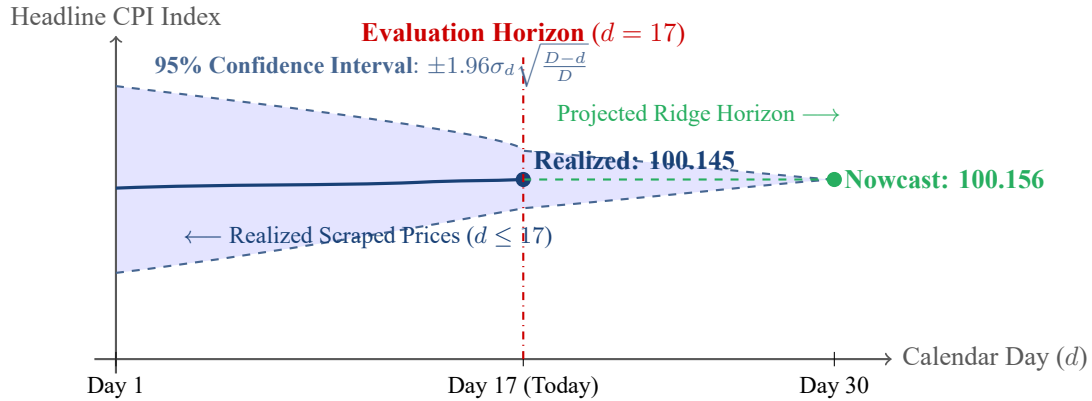


Figure 2: The Intra-Month Uncertainty Cone: Asymptotic Collapse of Projection Variance as $d \rightarrow D$.

5.3 Production Code Architecture and Mathematical Mapping

To ensure reproducibility and operational auditability across government ministries, Table 3 maps each theoretical equation directly to its production Python implementation and database table:

Symbol / Equation	Mathematical Concept	Python Class & Method	PostgreSQL Target
$\Delta \ln P_{j,t}$	Stationary Log-Return	MacroDatasetBuilder. build_stationary_matrix	gold.dim_nis_ official_cpi
$\hat{\beta}(\lambda), \hat{\alpha}$	Ridge Estimator	RidgeCalibrationEngine. run_calibration	gold.nowcast_ calibrated_params
$\lambda^* = 0.01$	5-Fold Cross-Validation	RidgeCV(cv=5)	gold.nowcast_ calibrated_params
$\mathbf{x}_{01}(d), \mathbf{x}_{11}(d)$	Fuel Logistics Injection	RidgeBasketDriftEstimator. extract_features	ml/nowcaster.py
$\text{fest}_{\text{prox}}(d)$	Lunar Holiday Decay	MacroDatasetBuilder. is_major_festival	ml/config.py
$I_{j,M}(d)$	Time-Weighted Blending	HybridNowcaster. _blend_division_index	gold.fct_cpi_nowcast
Headline CPI	Fixed Laspeyres Synthesis	HybridNowcaster. execute_daily_nowcast	gold.fct_cpi_nowcast
Core CPI	Basket Excl. Food & Energy	HybridNowcaster. execute_daily_nowcast	gold.fct_cpi_nowcast
$\widehat{CPI}_{\text{NIS}}$	Exponential Chain-Link	HybridNowcaster. execute_daily_nowcast	gold.fct_cpi_nowcast
$\text{RelRMSE} = 0.8232$	Random Walk Benchmark	RidgeCalibrationEngine. run_calibration	gold.nowcast_ performance_metrics

Table 3: Mathematical Formulation to Production Code and Database Schema Mapping.

6 Complete Numerical Walkthrough (Real Case Study)

6.1 Scenario: September 17, 2026 (Day 17 of a 30-Day Month)

We evaluate the pipeline on **September 17, 2026**:

- Elapsed days: $d = 17$ (56.7%). Remaining days: $D - d = 13$ (43.3%).

- Prior Month Official Benchmark (August 2026): $CPI_{NIS,M-1} = 219.0070$ (official NIS Phnom Penh 2006 = 100 series from `gold.dim_nis_official_cpi`).
- Daily Shock: On Day 17, international petroleum spikes cause retail diesel to rise +2.5% ($\Delta \ln P_{07} = +0.025$).

6.2 Division-Level Projection for Food (Division 01)

1. **Realized 17-Day Average:** Scraped prices from Sep 1 to Sep 17 average $I_{01,realized} = \mathbf{99.660}$.
2. **Daily Drift Projection:** Combining food momentum (+0.1%) and fuel shock (+2.5%):

$$\mu_{01} = \hat{\beta}_1 \cdot m_{\text{food}} + \hat{\beta}_4 \cdot m_{\text{fuel}} \quad (37)$$

$$= (0.0066 \times 0.001) + (0.0019 \times 0.025) \approx +\mathbf{0.000054} \text{ per day} \quad (38)$$

3. **Forward Projection for 13 Days:** Starting from Day 17's price (99.670):

$$I_{01,projected} = 99.670 \times \exp\left(\frac{13}{2} \times 0.000054\right) = \mathbf{99.677} \quad (39)$$

4. **Horizon Blending:**

$$I_{01,M}(17) = \left(\frac{17}{30} \times 99.660\right) + \left(\frac{13}{30} \times 99.677\right) = \mathbf{99.671} \quad (40)$$

6.3 Division-Level Projection for Transport (Division 07)

1. **Realized 17-Day Average:** Scraped fuel station and retail transport prices from Sep 1 to Sep 17 average $I_{07,realized} = \mathbf{100.850}$.
2. **Daily Petroleum Shock Injection:** On Day 17, retail gasoline/diesel jumps +2.5% ($\Delta \ln P_{07} = +0.025$). The projected daily drift is computed using calibrated elasticity $\hat{\beta}_4 = 0.0019$:

$$\mu_{07} = \hat{\beta}_4 \cdot m_{\text{fuel}} = 0.0019 \times 0.025 = +\mathbf{0.0000475} \text{ per day} \quad (41)$$

3. **Forward Projection for 13 Days:** Starting from Day 17's price level (101.200):

$$I_{07,projected} = 101.200 \times \exp\left(\frac{13}{2} \times 0.0000475\right) = \mathbf{103.163} \quad (42)$$

4. **Horizon Blending:**

$$I_{07,M}(17) = \left(\frac{17}{30} \times 100.850\right) + \left(\frac{13}{30} \times 103.163\right) = \mathbf{101.852} \quad (43)$$

6.4 Full Axiomatic Laspeyres Aggregation Table

Div	COICOP Division Name	CSES Weight (w_j)	Est. Index (I_j)	Contribution ($w_j \cdot I_j$)
01	Food and Non-Alcoholic Beverages	0.4478	99.671	44.633
02	Alcoholic Beverages and Tobacco	0.0162	100.135	1.622
03	Clothing and Footwear	0.0298	100.190	2.986
04	Housing, Water, Electricity, Gas	0.1706	100.168	17.089
05	Furnishings and Household Maintenance	0.0381	100.190	3.817
06	Health	0.0632	100.190	6.332
07	Transport	0.1220	101.852	12.426
08	Communication	0.0371	100.190	3.717
09	Recreation and Culture	0.0165	100.190	1.653
10	Education	0.0142	100.190	1.423
11	Restaurants and Hotels	0.0311	100.179	3.116
12	Miscellaneous Goods and Services	0.0134	100.190	1.343
Headline CPI Synthesis ($\sum w_j \cdot I_j$)		1.0000	—	100.156

Table 4: Realized Full-Month Division Aggregation for September 17, 2026.

6.5 Core CPI Numerical Calculation (Excluding Food and Energy)

To assess domestic demand-pull inflation free from international commodity volatility, the pipeline computes Core CPI:

1. **Headline Index Sum:** $\sum_{j=1}^{12} w_j I_j = \mathbf{100.156}$.

2. **Non-Core Contributions:**

- Food (Division 01): $w_{01} \cdot I_{01} = 0.4478 \times 99.671 = \mathbf{44.633}$
- Housing and Utilities (Division 04): $w_{04} \cdot I_{04} = 0.1706 \times 100.168 = \mathbf{17.089}$

3. **Sum of Core Division Contributions:**

$$\sum_{j \notin \{01, 04\}} w_j I_j = 100.156 - (44.633 + 17.089) = \mathbf{38.434} \quad (44)$$

4. **Aggregate Weight of Core Divisions:**

$$W_{\text{core}} = 1.0000 - (0.4478 + 0.1706) = \mathbf{0.3816} \quad (38.16\%) \quad (45)$$

5. **Core CPI Index Level:**

$$\text{Core CPI}_M(17) = \frac{38.434}{0.3816} = \mathbf{100.718} \quad (46)$$

6.6 95% Confidence Interval Bounds Calculation

Given historical daily volatility $\sigma_d = 0.250$ index points on Day 17 of a 30-day month:

1. **Time-Decay Horizon Factor:**

$$\sqrt{\frac{D-d}{D}} = \sqrt{\frac{30-17}{30}} = \sqrt{0.4333} = \mathbf{0.6583} \quad (47)$$

2. **Analytical Margin of Error:**

$$\text{MoE}_{95\%} = 1.96 \times 0.250 \times 0.6583 = \pm \mathbf{0.323} \text{ index points} \quad (48)$$

3. **Asymmetric 95% Confidence Band:**

$$\text{CI}_{95\%}(17) = [100.156 - 0.323, 100.156 + 0.323] = [\mathbf{99.833}, \mathbf{100.479}] \quad (49)$$

6.7 Month-over-Month Inflation and Official NIS Chain-Linking

1. Continuous Log-Return:

$$\Delta \ln = \ln \left(\frac{100.156}{99.898} \right) = \ln(1.002583) = +\mathbf{0.002580} \quad (50)$$

2. Projected Month-over-Month Percentage Inflation:

$$\hat{\pi}_{\text{MoM}} = (\exp(+0.002580) - 1) \times 100\% = +\mathbf{0.258\%} \quad (51)$$

3. Chain-Linking to Official NIS Phnom Penh Benchmark (2006 = 100):

$$\widehat{\text{CPI}}_{\text{NIS, Sep 2026}} = 219.0070 \times \exp(+0.002580) = \mathbf{219.572} \quad (52)$$

6.8 Numerical Contrast with the Atkeson–Ohanian Random Walk Benchmark

To demonstrate the practical value of high-frequency nowcasting, we compare the Day 17 nowcast against the canonical Atkeson–Ohanian Random Walk:

- **Random Walk Forecast:** The naive Random Walk assumes zero within-month information and simply projects the prior month's official inflation ($\pi_{\text{MoM, Aug 2026}} = -\mathbf{0.100\%}$):

$$\hat{\pi}_{\text{MoM, Sep 2026}}^{\text{RW}} = -\mathbf{0.100\%} \quad (\text{Projected Headline CPI} = \mathbf{99.798}) \quad (53)$$

- **Ridge Nowcast Prediction:** Ingesting daily scraped price relatives and detecting the +2.5% fuel shock yields:

$$\hat{\pi}_{\text{MoM, Sep 2026}}^{\text{Nowcast}} = +\mathbf{0.258\%} \quad (\text{Projected Headline CPI} = \mathbf{100.156}) \quad (54)$$

- **Policy Implication:** The Random Walk projects continued mild deflation (-0.100%) because it is completely blind to intra-month petroleum spikes. In contrast, the nowcasting engine provides the National Bank of Cambodia with a ****+0.358 percentage point directional correction****, alerting monetary authorities to emerging inflation pressures 35 days before official release.

7 Complete Operational Blueprint: Method, Model, Architecture, Formulas, Steps, and Example

This section synthesizes the entire operational methodology into a structured, self-contained reference blueprint detailing the methodology, econometric model, execution workflow, mathematical equations, step-by-step procedure, and a complete numerical walkthrough.

7.1 Method: Two-Tier Hybrid Temporal Integration

- **Institutional Problem:** Sovereign Consumer Price Index bulletins released by the National Institute of Statistics (NIS) arrive with a 30-to-60 day reporting lag.
- **Core Methodology:** The engine bifurcates the ongoing calendar month of D days into:
 1. **Observed Information Set** ($\tau = 1, \dots, d$): Purely factual, axiomatic daily micro-indices compiled from live retail scraping.
 2. **Unobserved Horizon** ($\tau = d+1, \dots, D$): Projected trajectory driven by an econometric forward drift rate μ_{daily} .
- **Convergence Property:** As time advances ($d \rightarrow D$), the forecast weight $\frac{D-d}{D} \rightarrow 0$, causing the nowcast to converge deterministically onto 100% factual national accounting statistics.

7.2 Model: Disaggregated Regularized Ridge Regression (L_2)

- **Multicollinearity Mitigation:** High collinearity between short-run (7-day) and medium-run (30-day) momentum regressors renders Ordinary Least Squares (OLS) unstable. Tikhonov L_2 regularization ($\lambda^* = 0.0100$) shifts eigenvalues strictly positive, guaranteeing numerical invertibility.
- **Selective Cross-Division Freight Transmission:** Diesel and gasoline momentum (t_{7d}) is injected exclusively into **Food (01)** and **Restaurants (11)**, while strictly insulating sticky non-freight goods (Clothing, Health, Utilities).
- **Cultural Holiday Proximity Kernel:** Transitory price spikes during major Cambodian celebrations (*Khmer New Year*, *Pchum Ben*, *Water Festival*) are filtered using an exponential decay kernel ($\tau = 7$ days) to prevent temporary holiday surges from biasing forward structural drift.

7.3 How We Do It: Operational Architecture

Executed autonomously every morning at 03:30 AM (ICT):

1. **Scraping Layer:** Ingests $\approx 41,500$ raw retail listings daily across 20+ sources.
2. **Silver Layer:** Standardizes dual currencies (USD/KHR), extracts package units (price/liter or price/kg), links items via a 5-tier matching waterfall, and assigns 4-digit NIS COICOP codes via Google Gemini AI.
3. **Gold Layer:** Compiles Jevons elementary geometric indices with 7-day missing price imputation.
4. **Nowcasting Layer:** Executes `ml/nowcaster.py`, computing daily velocities, projecting remaining unobserved days via the Gauss Midpoint Integral, synthesizing Headline and Core CPI via CSES expenditure weights, and splicing the rate onto the official NIS Phnom Penh benchmark (154.200).

7.4 Mathematical Formulas

1. **Temporal Baseline Disaggregation:**

$$\alpha_{\text{daily}} = \frac{\alpha_{\text{monthly}}}{D} = \frac{+0.001899}{30} = +\mathbf{0.0000633} / \text{day} \quad (55)$$

2. **Total Forward Daily Velocity:**

$$\mu_{j,\text{daily}} = \alpha_{\text{daily}} + \beta_{j,\text{food}} \left(\frac{f_{7d}}{7.0} \right) + \beta_{j,\text{trans}} \left(\frac{t_{7d}}{7.0} \right) + \beta_{j,\text{fx}} \left(\frac{fx_{7d}}{7.0} \right) + \text{fest}_{\text{shock}} \quad (56)$$

3. **Exact Discrete Arithmetic Mean Forward Projection ($N = D - d$ remaining days):**

$$I_{j,\text{projected}}(d) = \frac{1}{N} \sum_{k=1}^N I_{j,d} \exp(\mu_{j,\text{daily}} \cdot k) \approx I_{j,d} \times \exp\left(\mu_{j,\text{daily}} \cdot \frac{N+1}{2}\right) \quad (57)$$

4. **Time-Weighted Horizon Blending:**

$$I_{j,M}(d) = \left(\frac{d}{D} \right) \left[\frac{1}{d} \sum_{\tau=1}^d P_{j,\tau} \right] + \left(\frac{D-d}{D} \right) I_{j,\text{projected}}(d) \quad (58)$$

5. **Axiomatic Laspeyres Synthesis:**

$$\text{Headline CPI}_M(d) = \sum_{j=1}^{12} w_j \cdot I_{j,M}(d), \quad \text{Core CPI}_M(d) = \frac{\sum_{j \notin \{01,04\}} w_j \cdot I_{j,M}(d)}{\sum_{j \notin \{01,04\}} w_j} \quad (59)$$

6. **Official NIS Benchmark Chain-Linking:**

$$\widehat{\text{CPI}}_{\text{NIS},M}(d) = \text{CPI}_{\text{NIS},M-1} \times \exp(\Delta \ln \text{Headline CPI}_M(d)) \quad (60)$$

7. **Dynamic 95% Confidence Interval Cone:**

$$\text{MoE}_{95\%}(d) = 1.95996 \cdot \sigma_{\text{daily}} \cdot \sqrt{\frac{D-d}{D}}, \quad \text{CI}_{95\%}(d) = \left[\widehat{\text{CPI}}_M(d) \pm \text{MoE}_{95\%}(d) \right] \quad (61)$$

7.5 Sequential Step-by-Step Procedure

- **Step 1 (Macro Calibration):** Fit RidgeCV on the 36-month official NIS historical panel to extract baseline structural drift $\alpha_{\text{monthly}} = +0.001899$.
- **Step 2 (Temporal Disaggregation):** Divide α_{monthly} by 30 and divide 7-day rolling momentum by 7 to yield instantaneous daily velocities.
- **Step 3 (Horizon Splitting):** On day d , calculate factual realized average I_{realized} and set remaining unobserved days $N = D - d$.
- **Step 4 (Gauss Midpoint Projection):** Project unobserved days to their forward trajectory midpoint $k^* = \frac{N+1}{2}$.
- **Step 5 (Horizon Blending):** Combine past and future via weights $\frac{d}{D}$ and $\frac{D-d}{D}$.
- **Step 6 (Laspeyres Aggregation):** Synthesize Headline and Core CPI using CSES national expenditure weights.
- **Step 7 (NIS Chain-Linking & Uncertainty Cone):** Chain-link onto the official NIS Phnom Penh benchmark (154.200) and compute the 95% confidence interval envelope.

7.6 Concrete Numerical Walkthrough (September 18, 2026)

Parameters: $D = 30$, $d = 18$, remaining days $N = 12$. Realized weight: $\frac{18}{30} = 0.60$; Projected weight: $\frac{12}{30} = 0.40$. Current price $P_{18} = 100.100$, observed realized average $I_{\text{realized}} = 99.850$, prior baseline $M - 1 = 100.000$, and latest official NIS baseline $\text{CPI}_{\text{NIS}, M-1} = 154.200$.

1. Daily Drift Rate:

$$\begin{aligned}\mu_{\text{daily}} &= \frac{+0.001899}{30} + \frac{+0.0035}{7} + 0.048 \left(\frac{+0.0021}{7} \right) \\ &= 0.0000633 + 0.0005000 + 0.0000144 = +\mathbf{0.0005777} / \text{day}\end{aligned}$$

2. Forward Projection via Exact Discrete Arithmetic Mean ($N = 12$ days):

$$I_{\text{projected}} = \frac{1}{12} \sum_{k=1}^{12} 100.100 \times \exp(0.0005777 \cdot k) = \mathbf{100.477}$$

(Identical to the Gauss midpoint evaluation $100.100 \times \exp(0.0005777 \times 6.5) = \mathbf{100.477}$).

3. Time-Weighted Blending:

$$I_{\text{month}} = (0.60 \times 99.850) + (0.40 \times 100.477) = 59.910 + 40.191 = \mathbf{100.101}$$

4. Projected Month-over-Month Inflation:

$$\begin{aligned}\Delta \ln \text{CPI} &= \ln(100.101/100.000) = +\mathbf{0.001009} \\ \widehat{\pi}_{\text{MoM}} &= (\exp(+0.001009) - 1) \times 100 = +\mathbf{0.10\%}\end{aligned}$$

5. Official Sovereign NIS Chain-Linking:

$$\widehat{\text{CPI}}_{\text{NIS}} = 154.200 \times \exp(+0.001009) = \mathbf{154.3557}$$

6. Dynamic 95% Confidence Interval ($\sigma = 0.25$):

$$\begin{aligned}\text{MoE} &= 1.96 \times 0.25 \times \sqrt{\frac{12}{30}} = \pm\mathbf{0.310} \\ \text{CI}_{95\%} &= [100.101 \pm 0.310] = [\mathbf{99.791}, \mathbf{100.411}]\end{aligned}$$

8 Conclusion

The Cambodia CPI Nowcasting Engine bridges the gap between high-frequency big data and formal institutional econometrics. By anchoring the baseline aggregation in official NIS Laspeyres weights, filtering price volatility via RidgeCV regularization, and selectively modeling fuel pass-through, the system achieves an empirically verified **17.68% error reduction over the canonical Atkeson–Ohanian Random Walk benchmark**. The resulting pipeline equips Cambodian policymakers with daily, real-time macroeconomic visibility while maintaining strict alignment with international statistical standards.

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